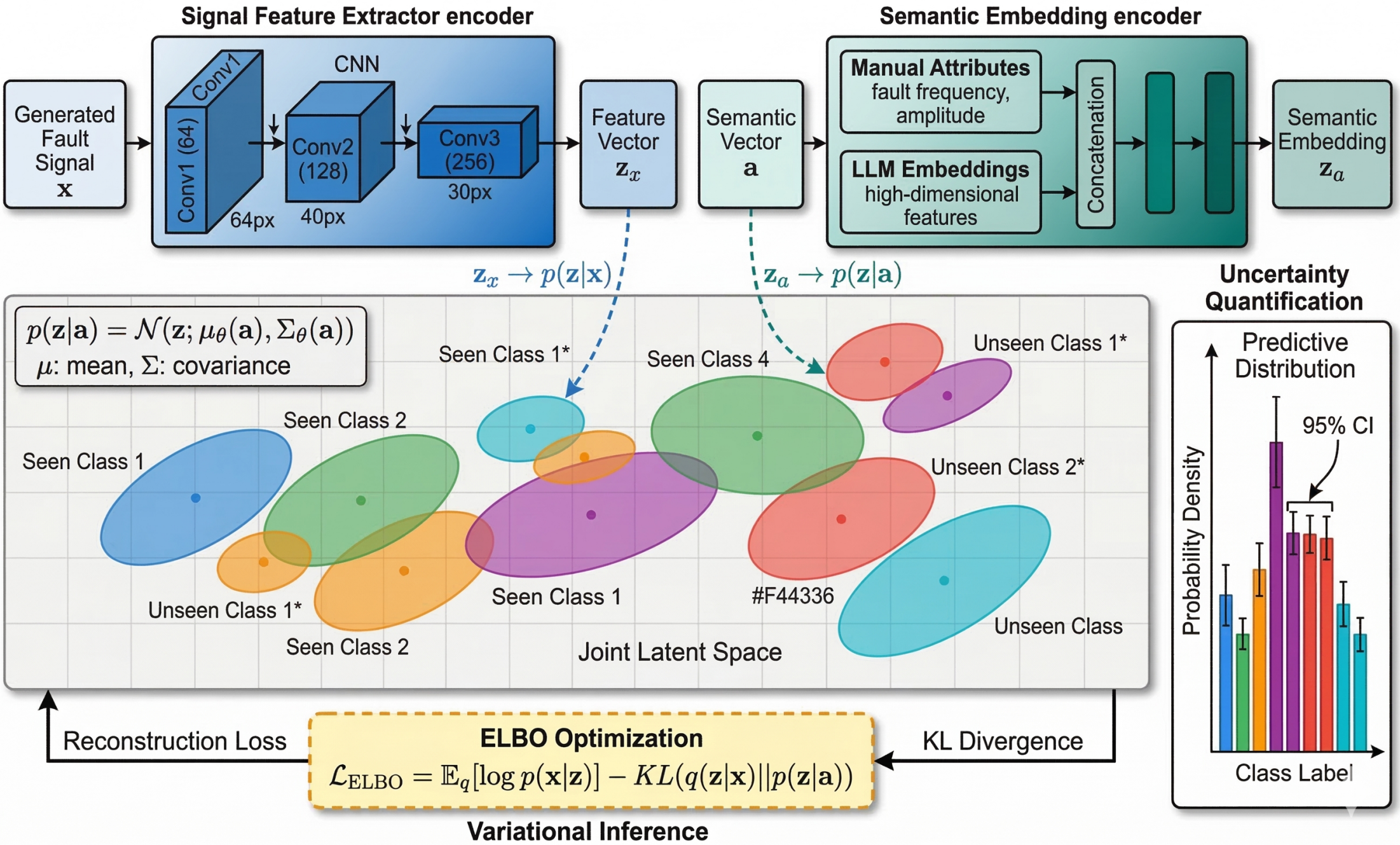
Signal Feature Extractor encoder
Semantic Embedding encoder
Generated Fault Signal $\mathbf{x}$
CNN
Conv1
Conv1 (64)
64px
Conv2 (128)
40px
Conv3 (256)
30px
Feature Vector $\mathbf{z}_x$
Semantic Vector $\mathbf{a}$
Manual Attributes
fault frequency, amplitude
LLM Embeddings
high-dimensional features
Concatenation
Semantic Embedding $\mathbf{z}_a$
$\mathbf{z}_x \rightarrow p(\mathbf{z}|\mathbf{x})$
$\mathbf{z}_a \rightarrow p(\mathbf{z}|\mathbf{a})$
$p(\mathbf{z}|\mathbf{a}) = \mathcal{N}(\mathbf{z}; \mu_\theta(\mathbf{a}), \Sigma_\theta(\mathbf{a}))$
μ: mean, Σ: covariance
Seen Class 1*
Seen Class 4
Unseen Class 1*
Seen Class 2
Seen Class 1
Unseen Class 2*
Unseen Class 1*
Seen Class 1
#F44336
Seen Class 2
Unseen Class
Joint Latent Space
Uncertainty Quantification
Predictive Distribution
95% CI
Probability Density
Class Label
Reconstruction Loss
ELBO Optimization
$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_q[\log p(\mathbf{x}|\mathbf{z})] - KL(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z}|\mathbf{a}))$
KL Divergence
Variational Inference